



Fiona Stanley Fremantle Hospitals Group

Policy and Procedure

Malnutrition

Reference #:FSFH-HW-PRO-0004

Scope

Site	Service/Department/Unit	Disciplines
Fiona Stanley Hospital Fremantle Hospital	Hospital Wide	Medical, Nursing, Allied Health
EXCLUSIONS	ICU; Maternity; Neonates; Burns; Mental Health; FH Day Admission Centre	

1. Introduction

Early identification of malnutrition and implementation of appropriate nutrition support reduces the risk of the patient's nutrition status deteriorating while in hospital.

The purpose of this document is to:

- identify patients who are at risk of malnutrition by Nurse-led malnutrition screening
- outline the diagnostic criteria for malnutrition
- outline the referral process for nutrition assessment and nutrition care planning

The information in this document is not applicable to the departments listed in the exclusions table above. In these areas consult with the ward Dietitian for management of patients with suspected malnutrition.

Terminology

Malnutrition	A state of nutrition in which an imbalance of energy, protein and other nutrients causes measurable adverse effects on body composition, function and clinical outcome. In clinical practice this term is predominantly used to describe undernutrition. This document focusses on undernutrition
Malnutrition Screening	A systematic approach for the early identification of patients who may be at risk of malnutrition
MST	Malnutrition Screening Tool. A screening tool to determine level of malnutrition risk in adult inpatients
PNST	Paediatric Nutrition Screening tool. A screening tool to determine level of malnutrition risk in paediatric inpatients

Nutrition Assessment	A comprehensive Dietitian-led approach to gathering pertinent data in order to evaluate nutritional status and identify nutrition-related problems in a patient
BMI	Body Mass Index. A measure to classify the body weight of adults. Calculated by dividing weight in kilograms by the square of height in meters
Chumlea equation	An equation for calculating height from knee height

2. Policy

- Malnutrition Screening Tool must be completed on all eligible patients admitted to FSFHG within 12 hours of admission.
- Paediatric Nutrition Screening Tool must be completed on all eligible paediatric patients admitted to FSH ward 3A within 12 hours of admission.

3. Adult malnutrition procedure

3.1. Screening on admission

The MST is completed on admission as part of the nursing admission form and recorded in the Digital Medical Record (DMR). Nursing action required from the MST score:

- MST Score 0-1
 - No immediate action. Repeat MST in 1 week using the digital medical record form
- MST score 2
 - Commence patient on High Energy High Protein Diet
 - Complete weekly weights
 - Commence patient on a 3-day food intake chart
 - Repeat MST in 1 week using the digital medical record form
 - Refer to Dietitian if patient consuming less than 50% of all meals
- MST score 3-5
 - Commence patient on High Energy High Protein Diet
 - Refer to Dietitian immediately for assessment as per usual ward practice. Place eReferral if after hours
 - Complete weekly weights
 - Commence patient on a 3-day food intake chart
 - Repeat MST in 1 week using the digital medical record form

3.2. MST re-screening

- Repeat the MST weekly for all inpatients. Use the MST form available in the digital medical record

3.3. Nutrition assessment (Dietitian led)

- Complete the Nutrition Care Process (NCP) and assess using one of the following validated tools as clinically indicated:
 - Subjective Global Assessment (SGA);
 - Patient Generated Subjective Global Assessment (PG-SGA);
- If unable to complete an SGA or PG-SGA, the ICD-10 criteria or the European Society for Parenteral and Enteral Nutrition (ESPEN) diagnostic criteria for malnutrition (EDC) can be considered as per Table 1.
- If the patient is not malnourished at the time of initial assessment, the Dietitian should still make comment on baseline nutrition status to allow for monitoring of change over time.
- Document the malnutrition diagnosis in the digital medical record using the “Nutrition Diagnosis Terminology” in Table 1

Table 1: Guide to Documentation of Malnutrition Diagnoses (Adults)

ICD10 Code	Nutrition Diagnosis Terminology	Supporting Documentation required
E43	Severe protein energy malnutrition	• SGA or PG-SGA C <u>OR</u> • BMI <18.5 kg/m² <u>OR</u> unintentional weight loss of >10% <u>AND</u> • Evidence of suboptimal intake resulting in severe loss of subcutaneous fat and/or severe muscle wasting
E44.0	Moderate protein energy malnutrition	• SGA or PG-SGA B <u>OR</u> • BMI <18.5 kg/m² <u>OR</u> unintentional weight loss of 5-9% <u>AND</u> • Evidence of suboptimal intake resulting in moderate loss of subcutaneous fat and/or moderate muscle wasting
E44.1	Mild protein energy malnutrition	• SGA or PG-SGA B <u>OR</u> • BMI <18.5 kg/m² <u>OR</u> unintentional weight loss of 5-9% <u>AND</u> • Evidence of suboptimal intake resulting in mild loss of subcutaneous fat and/or mild muscle wasting

E46	Unspecified protein energy malnutrition	• MST ≥ 2 <u>AND</u> • Unintentional weight loss (>10% indefinite time or >5% in last three months) <u>OR</u> • BMI $<20\text{kg/m}^2$ (if under 70 years of age) or BMI $<22\text{kg/m}^2$ (if over 70 years of age) <u>WITH</u> • Evidence of suboptimal intake resulting in loss of subcutaneous fat and/or muscle wasting
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- To ensure accurate coding and appropriate funding, the Dietician will record all malnutrition diagnoses in the digital medical record as either:
 - hospital acquired OR
 - pre-existing
- Discuss the outcomes of nutrition assessment with the clinical team and provide recommendations for treatment.

3.4. Nutrition intervention

A coordinated multi-disciplinary approach must be implemented when providing nutrition that adapts to the patient's individual needs and preferences. Nutrition management of malnutrition may include the following interventions:

- provision of high energy high protein diet
- dietary modifications using food-based strategies
- oral nutrition supplementation
- enteral feeding
- parenteral feeding
- education

4. Paediatric nutrition screening procedure

4.1. Screening on admission

The PNST is completed on admission as part of the nursing admission form and recorded in the digital medical record.

Action required from the PNST score:

- PNST Score 0-1
 - No immediate action required. Repeat screen in 1 week

- PNST score 2 or greater:
 - Refer immediately to Paediatric Dietitian for assessment as per usual ward practice. Place eReferral if after hours

4.2. Paediatric nutrition assessment (Dietitian led)

4.2.1. Complete weight, height or length measurements. Measurement of head circumference is completed at birth and 6-8 week.

- Weight is measured with children less than 1 year of age being bare, and children over 1 year wearing only a dry nappy or light clothing. Document what the child is wearing at the time of measurement.
- Height or length measurement is completed with a stadiometer or a length board/roll.
- Measurements of lower leg length or knee-heel length can be used for growth if accurate height or length cannot be obtained using standard methods. Height is estimated from lower leg length using the Chumlea equation for age.

4.2.2. Plot measurements on the growth chart in the medical record.

- Make corrections for prematurity for infants and children aged 0-2 years.

4.2.3. Document the malnutrition diagnosis in the digital medical record using the “Nutrition Diagnosis Terminology” in Table 2.

Table 2: Guide to Documentation of Malnutrition Diagnoses (Children)

ICD10 Code	Nutrition Diagnosis Terminology	Supporting Documentation required
E44.1	Mild protein energy malnutrition	0-2 years age • Weight for length from World Health Organisation (WHO) growth charts -1 to -1.9 z-score • Presence of stunting
		2-20 years • BMI from Centres for Disease Control (CDC) growth charts -1 to -1.9 z-score • Unintentional weight loss 5-7.4% • Presence of stunting

E44.0	Moderate protein energy malnutrition	0-2 years age - Weight for length from WHO growth charts -2 to -2.8 z-score, moderately wasted - Presence of stunting
		2-20 years - BMI from CDC growth charts -2 to -2.8 z-score, moderately wasted - Unintentional weight loss 7.5-9.9% - Presence of stunting
E43.0	Severe protein energy malnutrition	0-2 years age - Weight for length from WHO growth charts <-3 z-score, severely wasted - Presence of stunting
		2-20 years - BMI from CDC growth charts <-3 z-score, severely wasted - Unintentional weight loss >10% - Presence of stunting

4.3. Paediatric nutrition intervention

A coordinated multi-disciplinary approach must be implemented when providing nutrition that adapts to the patient's individual needs and preferences. Nutrition management of malnutrition in the paediatric population may include the following interventions:

- dietary modifications using food-based strategies
- provision of high energy high protein diet
- oral nutrition supplementation
- enteral feeding
- family education

5 Compliance/performance monitoring

The Nutrition and Hydration Committee will be responsible for monitoring compliance with this policy and procedure. Compliance will be evaluated via the FSFHG WECAN audit.

6 Related policy documents

- FSFHG:
 - Managing refeeding syndrome in adults (FSFH-ALLH-PRO-0005)
 - Inpatient Clinical Nutrition (FSH-HW-PRO-0009)

7 Related standards

- NSQHS Standards:
 - Clinical Governance
 - Comprehensive Care

8 Bibliography

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8. Evidence based practice guidelines for the nutritional management of malnutrition in adult patients across the continuum of care. (2009). *Nutrition & Dietetics*, 66(3), S1-S34.

9 Authorisation

EXECUTIVE SPONSOR: Director of Allied health					
Version	Date Issued	Compiled/Revised By	Committee/Consumer Group Consulted	Endorsed By	Revision due
1	08/2014			FSHEC	08/2017
2	06/2019	Head of Department, Dietetics; Senior Dietitian	FSFHG Nutrition and Hydration Committee; NMPGC	FSFHG Policy Committee	06/2022
3	07/2023	Senior Dietician; Dietetics HOD	FSFHG Nutrition & Hydration Committee; NMPGC	FSFHG Policy Committee	07/2026